MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC WITH 4PDT CENTER OFF 5 AMPERE AUXILIARY CONTACTS, 3PDT, 135 AMPERE CENTER OFF PRIMARY CONTACTS, TYPE II (UNSEALED), MECHANICALLY INTERLOCKED

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-6106.

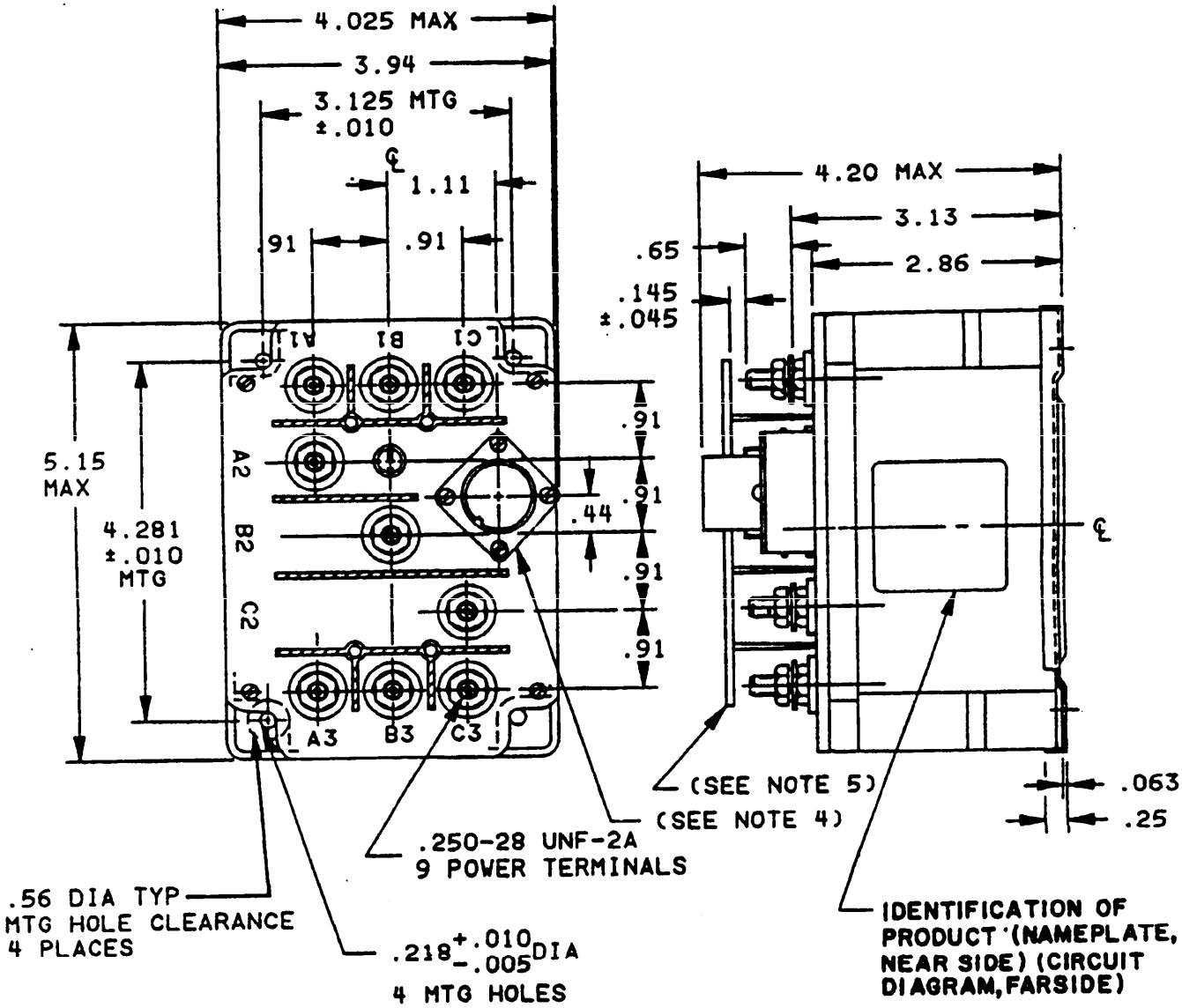
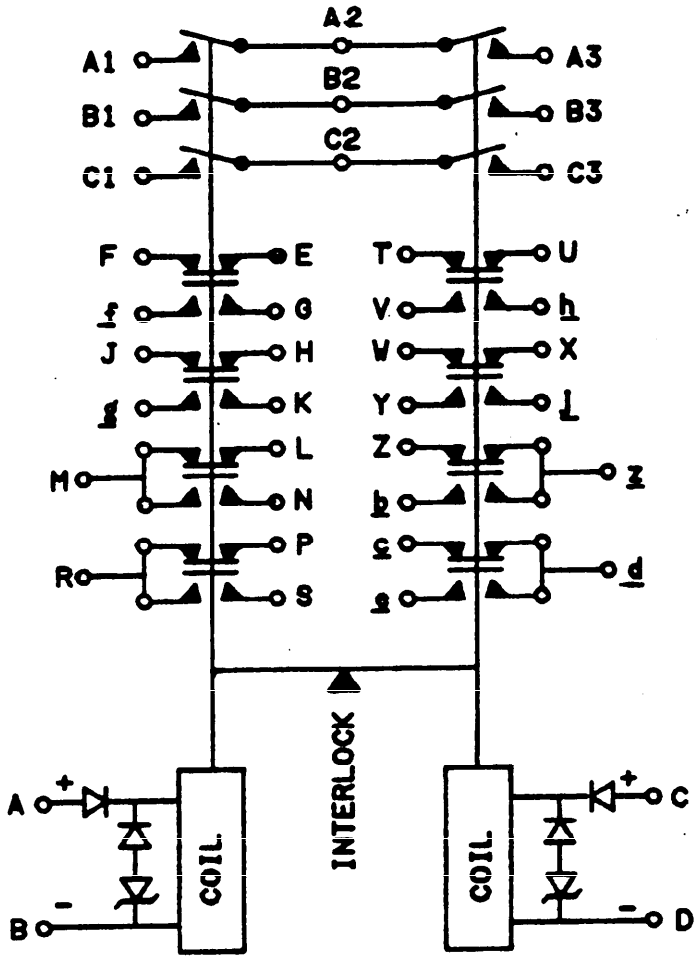


FIGURE 1. Dimensions and configuration.

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Circuit diagram (see note 6).

Inches	mm	Inches	mm
.005	0.13	0.91	23.1
.030	0.76	1.11	28.2
.045	1.14	2.86	72.6
.063	1.60	3.125	79.38
.145	3.68	3.13	79.5
.218	5.54	3.94	100.1
.250	6.35	4.025	102.24
.44	11.2	4.20	106.7
.56	14.2	4.281	108.74
.65	16.5	5.15	130.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.030 (0.76 mm).
4. Connector shall be a MIL-C-26482, class E, part number MS3112-E-18-32P.
5. Terminal barrier shall be polysulphone or polyetherkeytone.
6. Peak inverse voltage shall be 42 volts dc maximum.

FIGURE 1 Dimensions and configuration - Continued.

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REQUIREMENTS:

CONTACT DATA:

Load ratings:

TABLE I. Rated contact load (amperes per pole) case grounded.

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/			
		Main		Aux		Main		Aux		Main		Aux	
		NO	NC	NO	NC	400 Hz	50/60 Hz	400 Hz	60 Hz	400 Hz	50/60 Hz	400 Hz	60 Hz
Resistive	50	---	---	5	5	135	---	5	---	135	---	5	---
Inductive	50	---	---	3	3	135	---	3	---	135	---	3	---
Motor	50	---	---	---	---	80	---	---	---	80	---	---	---
Lamp	50	---	---	1.5	1.5	---	---	1.5	---	---	---	1.5	---
Transfer load	10	---	---	---	---	135	---	---	---	135	---	---	---
Mechanical life at reduce current	100	---	---	1.25	1.25	33.75	---	1.25	---	33.75	---	1.25	---
Intermediate current	50	13.5	13.5	Per MIL-R-6106									

1/ Absence of value indicates relay is not rated for 3 phase application.

Contact voltage drop:

Initial: 0.4 V maximum (including connector and lead resistance).

After life: 0.45 V maximum (including connector and lead resistance).

Contact bounce:

Primary contacts: 2 milliseconds maximum.

Auxiliary contacts: 4 milliseconds maximum.

COIL DATA: See table II.

Operate time: 60 milliseconds maximum with nominal coil voltage.

Release time: 50 milliseconds maximum from nominal coil voltage.

Transfer time: 10 milliseconds maximum.

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TABLE II. Dash numbers and characteristics.

Dash numbers M6106/52-	Coil data							
	Nominal voltage (V dc) 1/	Maximum voltage (V dc)	Res. Ω (ref. only) 2/ 3/	Max coil current (A) 3/		Pickup voltage (V dc)		
				Inrush 5/	Steady	Normal 4/	High temperature test	Continuous current test
001	28	30	10	3.5	1.5	15	18	22.5

1/ CAUTION: Use of any voltage less than nominal voltage will compromise the operation of the relay.
2/ DC coil resistance is not directly measurable at coil terminals.
3/ Measured at 20°C.
4/ Normal pickup voltage is applicable over the temperature range.
5/ Inrush prevails until contact closure is complete, at which time steady current prevails.

ELECTRICAL CHARACTERISTICS:

Insulation resistance:

Initial: 200 megohms minimum.

After life or environmental tests: 100 megohms minimum.

Dielectric withstanding voltage (sea level):	Initial V rms (60 Hz)	After life tests V rms (60 Hz)
Coil to case	1,250	1,000
Auxiliary contacts	1,250	1,000
All other points	1,800	1,350

Dielectric withstanding voltage (altitude):	50,000 FT V rms (60 Hz)
Coil to case	500
Auxiliary contacts	500
All other points	700

ENVIRONMENTAL CHARACTERISTICS:

Temperature range: -55°C to +71°C.

Maximum altitude rating: 50,000 feet.

Shock (specified pulse): MIL-STD-202, method 213, test condition J (30 g's with a duration of 11 milliseconds).

Vibration (sinusoidal): Applicable 0.06 double amplitude (5 to 60 hertz), 10 g's (60 hertz to 2000 hertz).

Contact chatter shall not exceed ten (10) microseconds, maximum for closed contacts and one (1) microsecond maximum for open contacts.

Salt spray (corrosion): Applicable.

Sand and dust: Applicable.

Explosion proof: Applicable.

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PHYSICAL DATA:

Terminal strength:

Stud terminals (threaded):

Pull force: 50 pounds (±1).

Torque: 75 inch-pounds (±1).

Weight: 3.25 pounds (1.475 kg) maximum.

Dimensions and configuration: See figure 1.

Seal: Gasket sealed and vented.

Identification of product: Applicable.

Part number: M6106/52-001.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodian:
Air Force - 85

Review activity:
DLA - ES

Preparing activity:
Air Force - 85

Agent:
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(Project 5945-F666)

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